

DATA CATALOG

NORTHWIND – TRADERS DATA WAREHOUSE PROJECT

Gold Layer Data Catalog

1. gold.dim_customers

Purpose: Stores descriptive customer information for customer-based analysis.

Grain: One row per customer.

Source: silver.customers

Column	Data type	Key type	Description
customer_key	BIGINT	Primary key	Dynamically generated customer dimension key using ROW_NUMBER().
customer_id	NVARCHAR(255)	Business key	Original customer identifier from the source system.
company_name	NVARCHAR(255)	—	Name of the customer's company.
contact_name	NVARCHAR(255)	—	Primary contact person for the customer.
contact_title	NVARCHAR(255)	—	Job title of the customer's contact person.
city	NVARCHAR(255)	—	City where the customer is located.
country	NVARCHAR(255)	—	Country where the customer is located.

2. gold.dim_employees

Purpose: Stores employee information and identifies each employee's manager.

Grain: One row per employee.

Source: silver.employees

Column	Data type	Key type	Description
employee_key	BIGINT	Primary key	Dynamically generated employee dimension key.
employee_id	INT	Business key	Original employee identifier.
employee_name	VARCHAR(255)	—	Full name of the employee.
employee_title	VARCHAR(255)	—	Employee's job title.
employee_city	VARCHAR(255)	—	City where the employee is based.
employee_country	VARCHAR(255)	—	Country where the employee is based.
manager_id	INT	Self-reference	Employee ID of the employee's manager.
manager_name	VARCHAR(255)	—	Name of the employee's manager.
manager_title	VARCHAR(255)	—	Job title of the employee's manager.

Transformation: A self-join is used to match manager_id with the corresponding employee record.

3. gold.dim_shippers

Purpose: Stores the shipping companies used to deliver orders.

Grain: One row per shipper.

Source: silver.shippers

Column	Data type	Key type	Description
shipper_key	BIGINT	Primary key	Dynamically generated shipper dimension key.
shipper_id	INT	Business key	Original shipper identifier.
company_name	VARCHAR(255)	—	Name of the shipping company.

4. gold.dim_products

Purpose: Stores product and category information for product-level analysis.

Grain: One row per product.

Sources: silver.products, silver.categories

Column	Data type	Key type	Description
product_key	BIGINT	Primary key	Dynamically generated product dimension key.
product_id	INT	Business key	Original product identifier.
product_name	NVARCHAR(50)	—	Name of the product.
quantity_per_unit	NVARCHAR(50)	—	Packaging or quantity description of the product.
unit_price	DECIMAL(10,2)	—	Standard unit price of the product.
discontinued	BIT or VARCHAR	—	Indicates whether the product has been discontinued.
category_id	INT	Business key	Original category identifier.
category_name	VARCHAR(MAX)	—	Name of the product category.
category_description	VARCHAR(MAX)	—	Description of the product category.

Transformation: Product records are enriched with category details using category_id.

5. gold.fact_order_line

Purpose: Stores order-line transactions and calculated sales and freight measures.

Grain: One row per order_id and product_id combination.

Sources: silver.orders, silver.order_details, and the four Gold dimension views.

Column	Data type	Key type	Description
order_key	BIGINT	Row identifier	Dynamically generated identifier for each order line.
order_id	INT	Degenerate dimension	Original order number retained in the fact view.
product_key	BIGINT	Foreign key	Links to gold.dim_products.
customer_key	BIGINT	Foreign key	Links to gold.dim_customers.
employee_key	BIGINT	Foreign key	Links to gold.dim_employees.
shipper_key	BIGINT	Foreign key	Links to gold.dim_shippers.
order_date	DATE	—	Date the order was placed.
required_date	DATE	—	Date by which the customer required delivery.
shipped_date	DATE	—	Date the order was shipped.
shipping_timeline	VARCHAR	Derived attribute	Classifies the shipment as early, on time, delayed, or unavailable.
unit_price	DECIMAL(10,2)	Measure	Unit price charged for the product line.
quantity	INT	Measure	Number of product units ordered.
discount	DECIMAL(3,2)	Measure	Discount rate applied to the order line.
sales_amount	DECIMAL(18,2)	Calculated measure	Net value of the product line after discount.
total_order_freight	DECIMAL(10,2)	Order-level measure	Total freight charged for the complete order.
allocated_freight	DECIMAL(12,2)	Calculated measure	Portion of order freight assigned to the product line.

Calculated fields

1. Sales Amount

$$Sales_amount = quantity \times unit_price \times (1 - discount)$$

2. Allocated freight

$$Allocated_freight = total_order_freight \times \frac{product_line_sales}{total_order_sales}$$

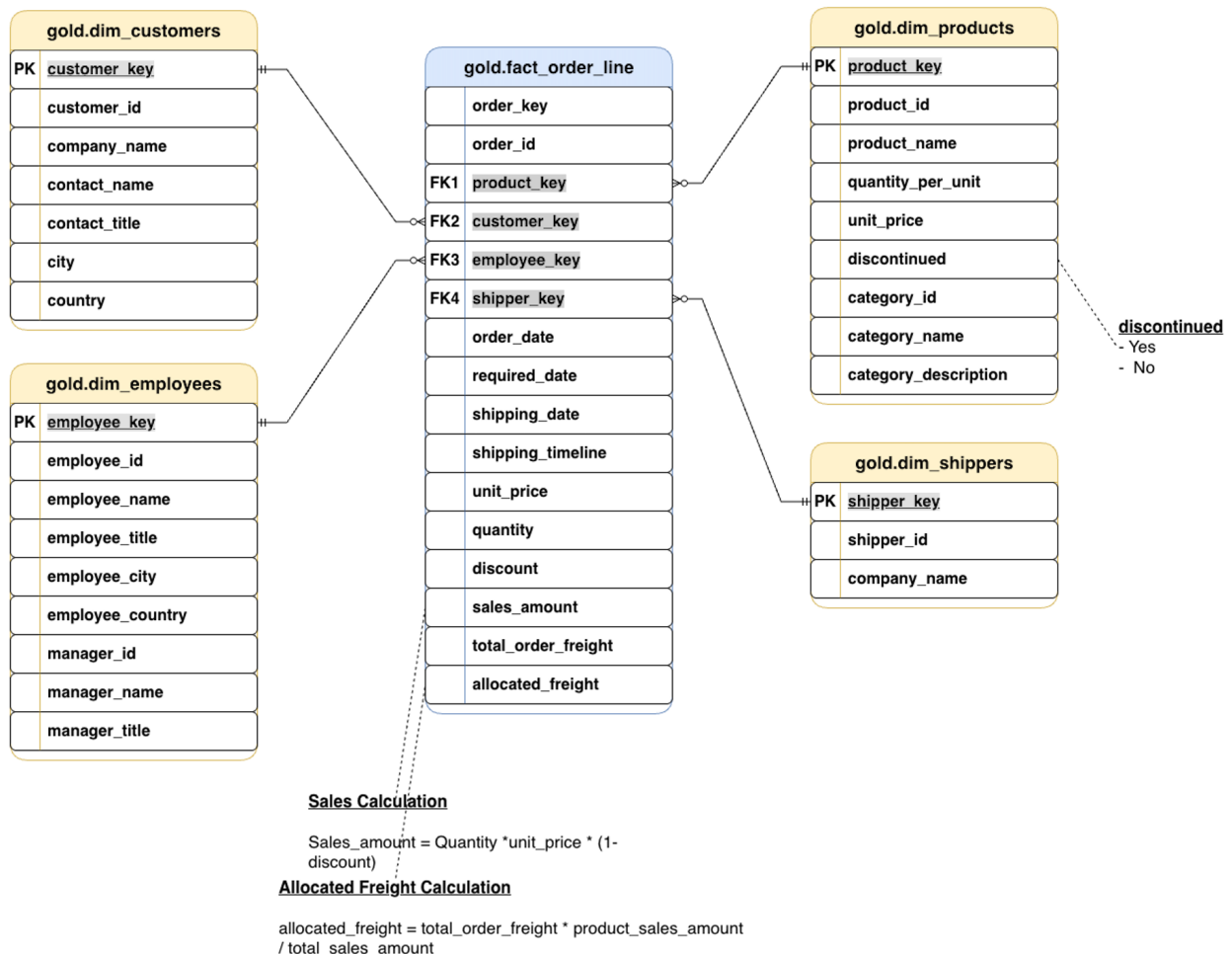
When total order sales (100% discount) are zero, freight is divided equally among the order lines.

3. Shipping timeline

Condition	Result
shipped_date < required_date	Shipped Early
shipped_date = required_date	Shipped on Time
shipped_date > required_date	Shipping Delayed
shipped_date IS NULL	N/A

Star-schema relationships

Fact-table column	Dimension-table column	Relationship
fact_order_line.customer_key	dim_customers.customer_key	Many order lines to one customer
fact_order_line.employee_key	dim_employees.employee_key	Many order lines to one employee
fact_order_line.shipper_key	dim_shippers.shipper_key	Many order lines to one shipper
fact_order_line.product_key	dim_products.product_key	Many order lines to one product



Important: Because the Gold layer uses views, surrogate keys created with ROW_NUMBER() are dynamically generated and may change when the underlying Silver-layer data changes.